A Data-Driven Digital Twin Framework for Type 1 Diabetes Management

Team C11

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1st Review Presentation

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Introduction

- Type 1 Diabetes (T1D) is a complex and continuously changing condition that requires effective understanding of individual patient dynamics.
- Patient conditions are influenced by multiple factors, including glucose, insulin, nutrition, physical activity, and sleep.
- The interactions between these factors make it challenging to represent a patient's condition using conventional single-source approaches.
- A Digital Twin provides a data-driven way to represent and study the evolving state of an individual patient.
- This project explores a Digital Twin framework for T1D management, with the aim of supporting better understanding, prediction, and simulation of patient dynamics.

Problem Statement

- Type 1 Diabetes is influenced by multiple interacting physiological and behavioral factors, making patient dynamics difficult to represent using a single variable or data source.
- Multimodal patient data are often heterogeneous, high-dimensional, and time-dependent, making it challenging to obtain a unified representation of the patient's condition.
- The changing nature of patient state over time makes it difficult to accurately model future states using conventional static approaches.
- There is a need for a framework that can learn patient-state dynamics from real-world longitudinal data while preserving the temporal nature of the observations.
- Therefore, the problem addressed in this project is to develop a data-driven Digital Twin framework capable of representing, learning, predicting, and simulating the evolving state of a T1D patient.

Objectives

- Integrate five T1D modalities — glucose, insulin, nutrition, physical activity, and sleep — into a unified learning framework.
- Develop five modality-specific GRU branches to capture the temporal patterns within each patient data stream.
- Fuse the learned multimodal representations into a 64-dimensional Unified Patient State.
- Learn patient-state dynamics through a data-driven Digital Twin for future-state prediction.
- Enable what-if simulation to explore how changes in the patient state may influence its future evolution.

Literature Review

S.No	Paper / Venue	Technique	Inference	Limitations
1	**GluNet: A Deep Learning Framework for Accurate Glucose Forecasting**	Personalized deep learning using glucose, meals and insulin	Effective personalized short-term glucose forecasting	Glucose-focused; no unified patient state
2	**Multi-step Ahead Predictive Model for Blood Glucose Concentrations of Type-1 Diabetic Patients**	Deep learning for multi-step blood glucose prediction	Supports multi-step BG forecasting using real T1D data	Focused on BG prediction; limited state representation
3	**Individualized Models for Glucose Prediction in Type 1 Diabetes prediction**	Physiological model, LSTM, GRU and TCN comparison	Highlights the importance of personalized temporal models	Primarily glucose prediction; no multimodal state fusion
4	**ReplayBG: A Digital Twin-Based Methodology to Identify a Personalized Model From Type 1 Diabetes**	Personalized glucose-insulin Digital Twin for therapy replay	Demonstrates Digital Twin-based treatment simulation	Glucose-insulin focused; limited multimodal representation

Literature Review

S.No	Paper / Venue	Technique	Inference	Limitations
5	**Design and In Silico Evaluation of an Exercise Decision Support System Using Digital Twin Models** (*Diabetes Technology & Therapeutics*, 2024)	Exercise-aware Digital Twin using heart rate, insulin and meal data to simulate glucose responses during aerobic and resistance exercise	Shows that Digital Twins can support personalized simulation of exercise-related glucose responses	Designed specifically for **exercise-related glucose management**; not a general multimodal patient-state model
6	**Platform for Precise, Personalised Glucose Forecasting** (*Computer Methods and Programs in Biomedicine*, 2024)	Wide-deep LSTM–GRU with attention using CGM, physical activity, carbohydrate intake and insulin	Shows that incorporating physical activity with metabolic data can improve personalized glucose forecasting	Still targets **glucose forecasting**; modalities are prediction inputs rather than a unified latent patient state
7	**Patient-specific Deep Offline Artificial Pancreas for Blood Glucose Regulation** (*2026*)	Systems-biology-informed neural network with deep reinforcement learning using glucose, carbohydrate intake and exercise intensity	Demonstrates potential for patient-specific computational modeling and automated glucose regulation	Focuses on **insulin dosing/control**; broader patient-state representation and multimodal state dynamics remain outside its scope

Literature Review

S.No	Paper / Venue	Technique	Inference	Limitations
8	**Mapping Data to Virtual Patients in Type 1 Diabetes** (*Control Engineering Practice*, 2020)	Personalizes the UVA/Padova T1D model using field-collected glucose, insulin, and meal data, with a variability component representing changes in insulin sensitivity	Demonstrates how real-world patient data can be mapped to personalized virtual patients for representing dynamic T1D behaviour	Uses a **physiological virtual-patient model** focused mainly on glucose–insulin dynamics rather than a learned multimodal patient-state representation

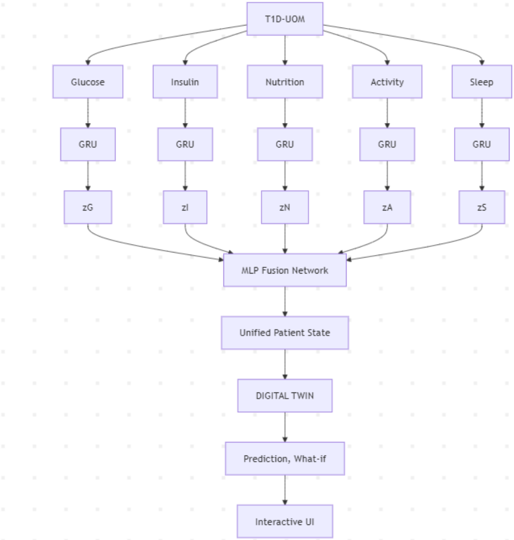
Research Gap Analysis

- **Fragmented multimodal modeling**: Existing T1D studies use glucose, insulin, nutrition, activity, or other factors, but a unified representation across multiple longitudinal modalities remains limited.
- **Prediction-centric approaches**: Most deep-learning methods primarily predict future blood glucose, rather than modeling the broader evolving patient state.
- **Narrow Digital Twin scope**: Existing T1D Digital Twins are largely glucose–insulin focused, with limited integration of diverse behavioral and physiological modalities.
- **Limited state-level simulation**: There is scope for a Digital Twin that learns patient-state dynamics and supports both future-state prediction and controlled what-if simulation from the learned state.

Dataset Description

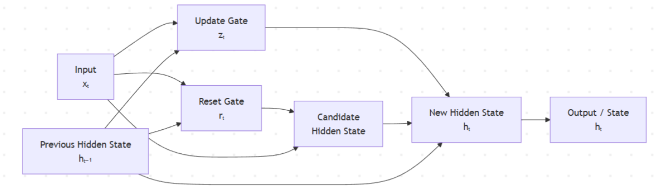
- **Dataset**: T1D-UOM longitudinal real-world Type 1 Diabetes data – University of Manchester (UoM)
- **Participants**: 11 — 9 training + 2 held-out. **Five modalities**: Glucose, Insulin, Nutrition, Physical Activity, Sleep Temporal data: Timestamped observations with different sampling densities across modalities.
  Evaluation: Held-out participants (UoM2401, UoM2405) were not used for training or optimization.
- Dataset Split
  Training — 9: UoM2301, UoM2302, UoM2304, UoM2305, UoM2306, UoM2307, UoM2308, UoM2309, UoM2313
  Held-out — 2: UoM2401, UoM2405

Architecture



What is GRU?

- Gated Recurrent Unit (GRU) is a recurrent neural network designed to learn patterns in sequential and time-dependent data.
- GRU uses reset and update gates to control how much past information is retained and how much new information is incorporated.
- It is well suited for longitudinal patient data, where the current patient state can depend on information observed at earlier time points.
- GRU provides a relatively simple and computationally efficient way to learn temporal dependencies compared with more complex recurrent architectures.



What is MLP Fusion

- MLP (Multi-Layer Perceptron) Fusion is a feed-forward neural network used to combine learned representations from multiple modalities.
- In our architecture, the outputs of the five GRU branches are brought together and provided to the fusion network.
- The MLP learns non-linear relationships and interactions across modalities, rather than treating each modality independently.
- The fused representation is transformed into a compact 64-dimensional Unified Patient State.
- This Unified Patient State provides the common representation that is subsequently used by the Digital Twin Dynamics for future-state prediction.

Digital Twin

- A **virtual, data-driven representation of a real patient** that reflects the patient's current diabetes-related state.
- Integrates **glucose, insulin, nutrition, physical activity, and sleep** information.
- Continuously updates its internal state from incoming patient data.

Twin Dynamics

The digital twin models how the patient's state changes over time:

$$
\begin{aligned}
\text{Current Unified Patient State:} &\quad S_t \in \mathbb{R}^{64} \\
\text{Predicted State Change:} &\quad \Delta S_t = f_\theta(S_t) \\
\text{Predicted Next State:} &\quad \hat{S}_{t+1} = S_t + \Delta S_t \\
\text{Combined Form:} &\quad \boxed{\hat{S}_{t+1} = S_t + f_\theta(S_t)} \\
\text{Residual Definition:} &\quad \Delta S_t = S_{t+1} - S_t \\
\text{Dimensions:} &\quad \Delta S_t \in \mathbb{R}^{64}, \qquad \hat{S}_{t+1} \in \mathbb{R}^{64}
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
S_t &= \text{Current Unified Patient State at time } t, \\
\Delta S_t &= \text{Predicted change in the patient state}, \\
f_\theta &= \text{TwinDynamics neural network}, \\
\theta &= \text{Learnable weights and biases of TwinDynamics},
\end{aligned}
}
$$

Methodology

1. Data Preparation: Process the five longitudinal modalities — glucose, insulin, nutrition, physical activity, and sleep — while preserving their original temporal observations.

2. Modality-Specific Processing: Convert each modality into its required numerical representation while handling modality-specific categorical information using the training-partition vocabulary.

3. Five Temporal GRU Branches: Use five independent GRU-based branches, one for each modality, to learn modality-specific temporal patterns from the longitudinal data.

4. Multimodal Fusion: Transform and combine the five learned representations using MLP-based fusion.

5. Unified Patient State: Generate a 64-dimensional Unified Patient State that provides a common representation of the patient's multimodal condition at each causal timestamp.

6. Temporal Transition Construction: Form consecutive state pairs from the Unified Patient State to represent patient-state evolution over time.

7. TwinDynamics: Train a neural transition model to learn the mapping from the current 64-dimensional state to the next patient state.

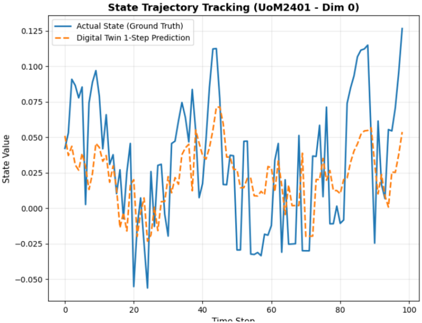
8. Digital Twin State: Initialize the Digital Twin from an observed Unified Patient State and evolve it using the learned TwinDynamics.

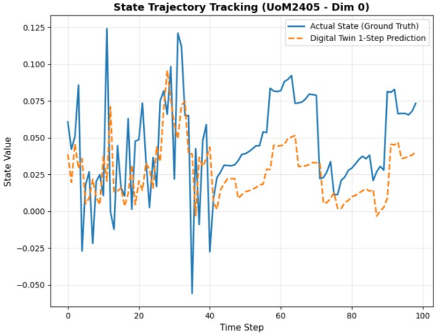
9. Future-State Prediction: Recursively apply the learned dynamics to generate simulated future states at multiple horizons (1, 5, 10, 30, and 60 steps).

Validation Strategy

- Train the dynamics model using 9 participants.
- Evaluate generalization on 2 completely held-out participants.
- Assess one-step prediction, recursive multi-step prediction, and what-if state effects.

Digital Twin – Held-Out Test Performance





Metric	**UoM2401**	**UoM2405**
Mean Squared Error (MSE)	**0.002668**	**0.002439**
Root Mean Squared Error (RMSE)	**0.0517**	**0.0494**
Mean Absolute Error (MAE)	**0.0409**	**0.0402**
Coefficient of Determination (R^2)	**0.9109**	**0.8509**

Future Work

- **What-If Simulation** to evaluate a wider range of clinically meaningful intervention scenarios.
- **Develop an Interactive Digital Twin UI** for visualizing patient trajectories, predictions, and what-if outcomes.
- **Add Explainable Predictions** to identify which modalities and patient-state factors contributed to each predicted change, improving transparency, confidence, and trust.
- **Develop Decision-Support Recommendations** based on predicted patient-state trajectories and simulated what-if outcomes.

Thank You